# Irfan Tayyab

Full-Stack AI Engineer

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## Education

**BS in Computer Science**, Virtual University of Pakistan

Final Year Project: an AI platform (FastAPI, LangChain, n8n) that routes students' queries from the web, email and WhatsApp. It uses Gemini with a Groq fallback, structured output and PII redaction, routing uncertain queries to a person.

## Work Experience

**Software Engineer**, AmentoTech Pvt Ltd — Nov 2025 – Oct 2026

- Architected a **configuration-driven form system** powering 8+ multi-step variations, with schema branching and per-step validation, making new flows quick to add and easy for the team to build on together.
- Set up a **Turborepo monorepo** with shared linting, formatting and pre-commit checks (lint-staged), giving the team and its AI tools one consistent codebase.
- Created shared, reusable components and complex layouts with React Portals, turning Figma designs into pixel-perfect UIs with AI-assisted tooling and **minimal designer-developer revision cycles**.
- Engineered a **personalized widget dashboard** on a drag-and-drop grid (dnd-kit), letting users rearrange, resize and reshape widgets while the layout holds on every screen size.
- Adopted **AI-driven development** with Claude and Cursor, using prompt engineering to streamline the work, alongside AI-powered pull request reviews and the team's custom skills plugin.

**Open Source Contributor**, Plumb MCP — Aug 2026

- Fixed a Figma **UI freeze** in Plumb MCP's plugin bridge by moving selection sync from push to on-demand pull, removing a full node-tree walk on every click; merged with all 955 tests passing (commit ↗).

**MERN Stack Engineer**, Sprouto Group — Mar 2025 – Nov 2025

- Developed and integrated a **RAG-pattern AI chatbot** in **7 working days**, parsing documents with LlamaParse into Supabase to ground its answers in retrieved sources.
- Led full-stack development within strict deadlines, delivering the corporate website in **1 month** with SEO-friendly server-side rendering and scroll-based parallax animations.
- Designed a typed **API integration layer** with centralized error handling, alongside reusable, type-safe React components.
- Managed the database, file storage and client-side server-state caching, cutting load times on slow connections.
- Migrated **1.9 GB** of data from MySQL on AWS RDS to PostgreSQL on Supabase, and **3 GB** of files from AWS S3 to Supabase Storage, with **zero data loss**.

**MERN Stack Developer**, CodeFlamme Pvt Ltd — Jul 2024 – Feb 2025

- Built a **real-time chat system** and live notifications with Socket.io (WebSockets).
- Implemented one-time payments and **auto-renewing subscriptions** with Stripe and PayPal, plus a referral system and its financial operations.
- Secured routes with role- and region-based authorization, plus Google, LinkedIn and Facebook OAuth sign-in.
- Localized the frontend with i18next and established a multi-language transactional email system on the backend.

## Skills

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| **AI & Automation** | LLM APIs (Gemini, Groq, Claude), LangChain, Structured Output, Model Fallback, RAG, Prompt Engineering, PII Redaction, NLP (Intent Detection, Entity Extraction, Text Classification), n8n Workflows, Gmail API, WhatsApp |
| **Languages** | Python, TypeScript, JavaScript, SQL |
| **Python** | FastAPI, SQLAlchemy 2 (async), asyncpg, Pydantic v2, Alembic, pytest, uv, Ruff, Pyright (strict) |
| **Frontend** | React.js, Next.js, Redux Toolkit, TanStack Query, React Hook Form, Zod, Tailwind CSS, Material UI, Figma |
| **Backend** | Node.js, NestJS, GraphQL (Apollo), REST APIs, WebSockets (Socket.io), JWT |
| **Databases** | PostgreSQL, MySQL, MongoDB, Prisma, Drizzle, Supabase, Firebase |
| **DevOps & Testing** | Docker, Docker Compose, Caddy, Oracle Cloud, Vercel, GitHub Actions (CI), Git, Jest/Vitest, React Testing Library |